sotFS

A Graph-Structured Filesystem with
DPO Rewriting, Bounded Treewidth, and
Capability-Gated Deceptive Projections

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Abstract

Filesystem metadata is inherently a graph: inodes link to directories, capabilities reference files, hard links create shared substructure, and version histories form DAGs. Yet every major filesystem treats metadata as flat tables or implicit trees, losing the graph structure that could enable powerful queries, anomaly detection, and formal reasoning about correctness.

SOTFS makes this graph structure explicit. It represents filesystem metadata as a *typed graph* with six node types and six edge types, and defines POSIX operations as *Double Pushout (DPO) graph rewriting rules*—algebraic transformations with mathematically precise preconditions, postconditions, and gluing constraints.

This formalization yields three structural properties unavailable in conventional filesystems: (1) **bounded treewidth**, enabling any MSO-definable query to execute in *linear time* via Courcelle's theorem; (2) **Ollivier–Ricci curvature monitoring**, detecting structural anomalies in real time; and (3) **deceptive subgraph views**, where different security domains see different projections of the filesystem graph.

SOTFS is designed for SOTOS, a capability-based microkernel written in Rust. It runs as a userspace server domain, communicating via IPC and persisting state through the SOT exokernel's transactional Secure Object primitives. A four-layer verification plan—TLA+ model checking, crash refinement, Rust typestate, and Coq/Perennial proofs—provides graduated formal assurance.

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1 Introduction and Motivation

1.1 The Problem: Filesystems as Implicit Graphs

Every filesystem manages a graph. A directory tree is trivially a graph. Hard links make it a DAG. Symbolic links add arbitrary edges. Mount points graft subtrees. Access control lists associate permission sets with inodes. Snapshots create version DAGs. Transactions group operations into atomic units referencing multiple nodes.

Yet no production filesystem treats this graph as a first-class data structure:

- **ext4** stores metadata in superblocks, group descriptors, inode tables, and directory entry blocks—flat structures connected by implicit integer references (inode numbers).
- **ZFS** uses a Merkle-like tree of block pointers where the graph structure is implicit in pointer chains.
- **Btrfs** uses B-trees with composite keys encoding path information, with backreferences for deduplication.
- **HAMMER2** uses a radix tree with embedded block references and snapshot chains forming implicit DAGs.

This implicit representation has three consequences: (1) queries are slow (graph queries forced through non-graph interfaces), (2) formal reasoning is hard (invariants must be reconstructed from code), and (3) security monitoring is blind (namespace structure must be inferred from syscall traces).

1.2 Thesis Statement

> *Filesystem metadata is a typed graph. By formalizing POSIX operations as DPO graph rewriting rules and maintaining the metadata graph as an explicit first-class data structure, we can prove that operations preserve structural invariants by verifying the DPO rules once, enable MSO-definable security queries in linear time, detect structural anomalies via Ollivier–Ricci curvature, and implement capability-gated graph projections for cyber deception.*

1.3 Contributions

1. **Type Graph $\mathcal{TG}$:** A formal definition of filesystem metadata as a typed graph with six node types and six edge types, with precise local and global invariants (§4).
2. **DPO Rule Set:** A complete specification of core POSIX operations as DPO graph rewriting rules with explicit gluing conditions (§5).
3. **Treewidth Preservation:** A proof that the DPO rules preserve bounded treewidth, enabling linear-time MSO queries via Courcelle's theorem (§7).
4. **Curvature Monitor:** An incremental Ollivier–Ricci curvature computation scheme for real-time anomaly detection (§8).
5. **Deceptive Projections:** Capability-gated subgraph views formalized as a graph endofunctor (§9).
6. **gtxn Protocol:** A graph-level transaction manager composing the SOT kernel's tiered transactions with graph-specific invariant checking (§6).

2 Background and Related Work

2.1 Double Pushout Graph Rewriting

DPO graph rewriting, developed by Ehrig, Pfender, and Schneider (1973) and formalized in Ehrig et al. [1], is a categorical approach to graph transformation. A **DPO rule** $\rho = (L \xleftarrow{l} K \xrightarrow{r} R)$ consists of a left-hand side L (pattern to match), an interface K (preserved portion), and a right-hand side R (replacement).

Application of ρ to a host graph G via a match $m\colon L \to G$ produces a result H through two pushout squares:

$$\begin{tikzcd} L \arrow[d, "m"'] & K \arrow[l, "l"'] \arrow[r, "r"] \arrow[d, "d"] & R \arrow[d, "m^*"] \\ G & D \arrow[l, "l^*"] \arrow[r, "r^*"'] & H \end{tikzcd} \tag{1}$$

The **gluing conditions** ensure applicability:

1. *Identification condition:* m must not identify elements in $L \setminus l(K)$.
2. *Dangling condition:* Removing $m(L \setminus K)$ must not leave edges with missing endpoints.

2.2 Courcelle's Theorem and Bounded Treewidth

Treewidth [11] measures how "tree-like" a graph is. A tree has treewidth 1; a complete graph K_n has treewidth $n - 1$.

Theorem 2.1 (Courcelle, 1990 [2])**.** Every graph property definable in Monadic Second-Order logic (MSO_2) can be decided in time $O(f(|\varphi|, k) \cdot n)$ on graphs of treewidth $\leq k$, where $n = |V|$.

MSO_2 can express reachability, cycle detection, k-colorability, dominating sets, and many security-relevant properties.

2.3 Ollivier–Ricci Curvature on Graphs

Ollivier [3] defined Ricci curvature for metric spaces via optimal transport. Lin, Lu, and Yau [4] adapted it to graphs:

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{2}$$

where W_1 is the Wasserstein-1 distance between probability measures μ_u, μ_v on the neighborhoods of u and v.

2.4 Comparison with Existing Systems

Table 1: Feature comparison of SOTFS with production filesystems.

Feature	**ext4**	**ZFS**	**Btrfs**	**HAMMER2**	**SquirrelFS**	**sotFS**
Metadata model	tables	Merkle	B-trees	radix	Rust types	**typed graph**
Formal ops	—	—	—	—	typestate	**DPO rules**
Treewidth bound	—	—	—	—	—	$\leq k$
MSO queries	—	—	—	—	—	$O(n)$
Anomaly detect.	—	—	—	—	—	**Ollivier–Ricci**
Cap. integration	DAC	DAC	DAC	DAC	—	**graph edges**
Deception	—	—	—	—	—	**subgraph views**
Verification	tested	tested	tested	tested	typestate	**TLA+→Coq**

3 Architecture Overview

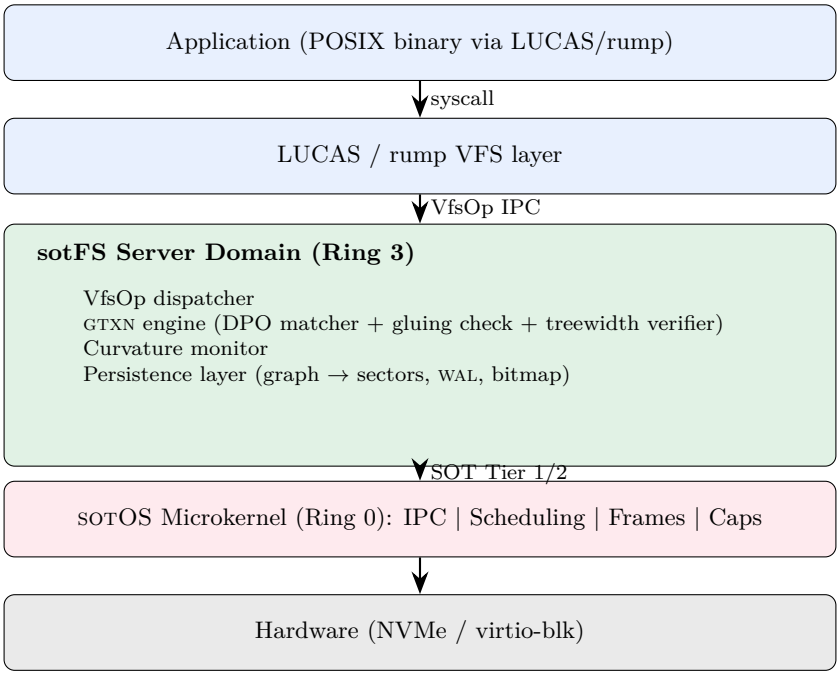

Figure 1: SOTFS within the SOTOS stack.

The SOTFS server domain contains five components: (1) VfsOp Dispatcher, (2) GTXN Engine, (3) Type Graph Store, (4) Curvature Monitor, and (5) Persistence Layer (extending the existing `ObjectStore` layout).

4 Type Graph Formalization

Definition 4.1 (Type Graph)**.** The SOTFS type graph is a tuple

$$\mathcal{TG} = (V, E, \mathrm{src}, \mathrm{tgt}, \tau_V, \tau_E, \mathrm{attr})$$

where V is a finite set of nodes, E is a finite set of edges, $\mathrm{src}, \mathrm{tgt} \colon E \to V$ map edges to endpoints, $\tau_V \colon V \to \texttt{NodeType}$ and $\tau_E \colon E \to \texttt{EdgeType}$ assign types, and $\mathrm{attr} \colon V \cup E \to \texttt{Attributes}$ assigns typed attribute records.

The type sets are:

$$\texttt{NodeType} = \{\texttt{Inode}, \texttt{Directory}, \texttt{Capability}, \texttt{Transaction}, \texttt{Version}, \texttt{Block}\}$$
$$\texttt{EdgeType} = \{\texttt{contains}, \texttt{grants}, \texttt{delegates}, \texttt{derivesFrom}, \texttt{supersedes}, \texttt{pointsTo}\}$$

4.1 Node Types

Each node type carries typed attributes:

Node Type	Key Attributes	Maps To (existing)
`Inode`	id, vtype, perms, uid, gid, size, link_count, timestamps	`Vnode` in `vnode.rs`
`Directory`	id	Paired with `Inode{vtype=Directory}`
`Capability`	cap_id, rights, epoch	`CapEntry` in `cap/table.rs`
`Transaction`	txn_id, tier, state, read_set, write_set	State machine in `sot_transactions.tla`
`Version`	version_id, timestamp, root_oid	`SnapMeta` in `layout.rs`
`Block`	block_id, sector_start, sector_count, refcount	`DirEntry` sectors in `layout.rs`

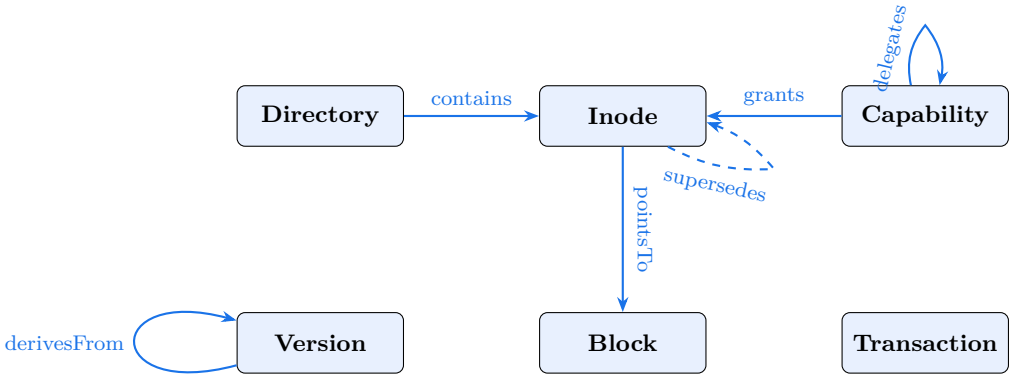

Figure 2: Node types and edge types of the SOTFS type graph $\mathcal{TG}$.

4.2 Edge Types and Constraints

Edge Type	Source → Target	Attribute	Constraint
`contains`	Directory → Inode	name: String	Unique name per Directory
`grants`	Capability → Inode	rights: Rights	Monotonic via delegation
`delegates`	Capability → Capability	—	CDT parent–child (forest)
`derivesFrom`	Version → Version	—	DAG (no cycles)
`supersedes`	Inode → Inode	—	Acyclic
`pointsTo`	Inode → Block	offset: u64	Non-overlapping extents

4.3 Invariants

4.3.1 Local Invariants

Invariant 4.1 (Link Count)**.** For every Inode i: $\text{link_count}(i) = |\{e \in E : \tau_E(e) = \texttt{contains} \wedge \text{tgt}(e) = i \wedge \text{attr}(e).\text{name} \neq \text{“.”}\}|$.
Invariant 4.2 (Directory Self-Reference)**.** Every Directory node d has exactly one outgoing `contains` edge with name = “.” pointing to its paired Inode.
Invariant 4.3 (Unique Names)**.** For every Directory d, the names on outgoing `contains` edges form a set (no duplicates).
Invariant 4.4 (Capability Exactness)**.** Every Capability node c has exactly one outgoing `grants` edge.
Invariant 4.5 (Block Refcount)**.** For every Block b: $\text{refcount}(b) = |\{e \in E : \tau_E(e) = \texttt{pointsTo} \wedge \text{tgt}(e) = b\}|$.
Invariant 4.6 (Non-Overlapping Sectors)**.** For any two Block nodes $b_1 \neq b_2$, their sector ranges are disjoint.

4.3.2 Global Invariants

Invariant 4.7 (No Orphan Inodes)**.** Every Inode with link_count > 0 is reachable from the root Inode via `contains` edges.
Invariant 4.8 (No Dangling Edges)**.** $\forall e \in E : \text{src}(e) \in V \wedge \text{tgt}(e) \in V$.
Invariant 4.9 (Capability Monotonicity)**.** For every path $c_0 \xrightarrow{\texttt{delegates}} c_1 \xrightarrow{\texttt{delegates}} \cdots \xrightarrow{\texttt{delegates}} c_n$: $\text{rights}(c_n) \subseteq \text{rights}(c_{n-1}) \subseteq \cdots \subseteq \text{rights}(c_0)$.
Invariant 4.10 (No Directory Cycles)**.** The `contains` subgraph restricted to `Directory` → `Inode{vtype=Directory}` edges (excluding “`.`” and “`..`”) is acyclic.
Invariant 4.11 (Treewidth Bound)**.** $\text{tw}(\mathcal{TG}) \leq k$ for a configurable constant k (see Theorem 7.1).

5 DPO Rules for POSIX Operations

We specify 7 mutating DPO rules. Read-only operations (`OPEN`, `READ`, `CLOSE`, `STAT`) are graph morphism queries—they do not mutate $\mathcal{TG}$.

5.1 CREATE(d, name, vtype)

Creates a new file or directory entry.

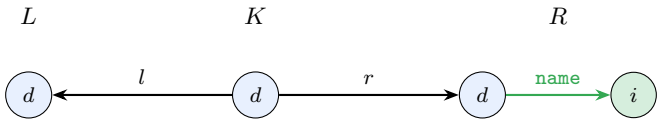

Figure 3: DPO rule for CREATE (file). Green = added.

Gluing conditions: (GC1) No existing `contains` edge from $m(d)$ with this name. (GC2) The match is unique.

Invariant preservation: Invariant 4.7: i is reachable from root via d. Invariant 4.1: link_count(i) = 1. Invariant 4.11: adding a leaf to a tree preserves treewidth. □

5.2 MKDIR(d, name)

Creates a new directory: adds Inode (vtype = `Directory`), Directory node, and three `contains` edges (`name`, "`.`", "`..`").

Tier: SOT Tier 1 (single Directory SO).

5.3 RMDIR(d, name)

Removes an empty directory. L includes the target Directory, its Inode, and all three `contains` edges. K retains only the parent.

Gluing: Target must be empty (only "`.`" and "`..`" edges).

5.4 LINK(d, name, i_{target})

Creates a hard link—a new `contains` edge to an existing Inode. Side effect: link_count(i_{target}) += 1.

Gluing: (GC1) vtype(i_{target}) $\neq$ `Directory`. (GC2) link_count $<$ `LINK_MAX` (critical for treewidth bound).

5.5 UNLINK(d, name)

Removes a `contains` edge and decrements link_count. If link_count reaches 0, garbage-collects the Inode and its Block nodes.

5.6 RENAME

The most complex rule, with four cases:

Case	Same dir?	Target exists?
A	yes	no
B	yes	yes (replace)
C	no	no
D	no	yes (replace)

Cases C and D touch two Directory SOs, requiring SOT Tier 2 (2PC). Case B/D adds a `supersedes` edge and decrements the replaced Inode's link_count.

Gluing (GC-RENAME-2): If the source is a directory and the destination is a descendant of the source, the rename would create a cycle—reject.

5.7 WRITE(i, offset, data)

Updates file data by adding or replacing Block nodes and `pointsTo` edges. CoW: if a Block's refcount > 1, allocate a new block.

5.8 Treewidth Impact per Rule

Rule	Nodes added	Edges added	tw impact
CREATE (file)	1 Inode	1 `contains`	$+0$
MKDIR	1 Inode + 1 Dir	3 `contains`	$+0$
RMDIR	-1 Inode -1 Dir	-3 `contains`	≤ 0
LINK	0	1 `contains`	0 to $+1$
UNLINK	0 to -1	-1 `contains`	≤ 0
RENAME	0	net 0	$+0$
WRITE	0–1 Block	0–1 `pointsTo`	$+0$

Only LINK can increase treewidth, bounded by `LINK_MAX`.

6 Capabilities and Transactions

6.1 Capabilities as Typed Graph Edges

Capabilities are nodes in $\mathcal{TG}$ connected by two edge types:

- `grants`(c, i): authorizes access to Inode i with rights rights(c).
- `delegates`$(c_{\text{parent}}, c_{\text{child}})$: represents CDT derivation with rights$(c_{\text{child}}) \subseteq$ rights(c_{parent}).

Capability attenuation is edge relabeling in the `grants` subgraph: the monotonicity invariant (4.9) is a statement about non-increasing labeling along paths in the `delegates` subgraph.

6.2 Graph Transactions (gtxn)

A GTXN wraps DPO rule applications into an atomic unit (ADR-001):

1. Verify DPO gluing conditions on host graph G.
2. Verify treewidth bound preserved.
3. Begin SOT Tier 1 or Tier 2 transaction.
4. Apply $L \to K \to R$ morphisms.
5. Verify post-conditions (global invariants).
6. **Commit:** persist via WAL, update curvature, emit provenance.
7. **Rollback:** SOT abort, discard in-memory changes.

DPO Rule	SOs touched	SOT Tier
CREATE, MKDIR, RMDIR	1 Dir + 1 Inode	Tier 1
LINK, UNLINK	1 Dir + 1 Inode	Tier 1
WRITE	1 Inode + 0–n Blocks	Tier 1
RENAME (same dir)	1 Dir + 1–2 Inodes	Tier 1
RENAME (cross-dir)	2 Dirs + 1–2 Inodes	**Tier 2 (2pc)**

7 Bounded Treewidth and MSO Queries

Theorem 7.1 (Treewidth Bound)**.** Let $\mathcal{TG}$ be a SOTFS type graph produced by any finite sequence of DPO rule applications from the CREATE-ROOT initial graph. If the hard link count per inode is bounded by `LINK_MAX`, then

$$\mathrm{tw}(\mathcal{TG}) \leq \texttt{LINK_MAX} + O(1).$$

Proof sketch. Decompose $\mathcal{TG}$ by edge type:

1. **`contains`** subgraph: without hard links, a forest (tw = 1). Each hard link adds one back-edge, increasing tw by at most 1. With $\leq$ `LINK_MAX` links per inode: tw(`contains`) $\leq$ `LINK_MAX`+1.
2. **`grants`**: bipartite, degree 1 on Capability side. tw = 1.
3. **`delegates`**: forest (CDT). tw = 1.
4. **`derivesFrom`**: DAG with bounded in-degree. tw ≤ 2.
5. **`pointsTo`**: stars from Inodes to Blocks. tw = 1.
6. **`supersedes`**: acyclic, out-degree ≤ 1. tw = 1.

The `contains` subgraph dominates. By the tree decomposition merging lemma [12]: $\mathrm{tw}(\mathcal{TG}) \leq$ `LINK_MAX` $+ O(1)$. □

Corollary 7.2. For filesystems without hard links, $\mathrm{tw}(\mathcal{TG}) \leq 6$.

MSO query examples (all $O(n)$ when tw is bounded):

Query	mso_2 Description
Q1: Can domain D read file F?	Reachability in `delegates`/`grants` subgraph
Q2: Cycle in directory structure?	Cycle detection in `contains` subgraph
Q3: Orphan inodes?	$\exists\, i\colon \neg\mathrm{reachable}(\mathrm{root}, i) \wedge \mathrm{link_count}(i) > 0$
Q4: Files derived from version V?	Transitive closure in `derivesFrom` subgraph

8 Ollivier–Ricci Curvature Monitor

We compute Ollivier–Ricci curvature $\kappa(u, v)$ (Eq. 2) with lazy random walk measure $\mu_x(z) = \alpha \cdot \mathbf{1}_{z=x} + \frac{1-\alpha}{\deg(x)} \cdot \mathbf{1}_{z \in N(x)}$, $\alpha = 0.5$.

8.1 Anomaly Signatures

Attack Pattern	Curvature Signature
Mass file creation (zip bomb)	Positive κ spike on victim directory edge
Symbolic link bomb	Strong positive κ on target inode's edges
Deep directory chain	Sustained $\kappa \approx -1.0$ (maximally tree-like)
Directory structure manipulation	Sudden κ change on previously stable edges

8.2 Incremental Computation

Each DPO rule changes $O(1)$ edges. We recompute κ only for changed edges and their 1-hop neighbors ($|\Delta^*| \leq 20$ for most operations). Amortized cost: $O(d_{\max}^2)$ per rule application, where $d_{\max}$ is the maximum degree of touched nodes.

9 Deceptive Subgraph Views

Definition 9.1 (Graph Projection)**.** A projection π for domain D is a typed graph morphism $\pi\colon \mathcal{TG}|_D \to \mathcal{TG}_D$, where $\mathcal{TG}|_D$ is the subgraph reachable from D's root capability and $\mathcal{TG}_D$ is the visible type graph. π may omit, add, or relabel nodes and edges.

We extend the existing deception functor $D\colon \mathbf{SO} \to \mathbf{SO}$ to a graph endofunctor:

$$D_{\mathrm{FS}}\colon \mathbf{TGraph} \to \mathbf{TGraph}$$

The four interposition policies map to functor behavior: **Passthrough** = identity, **Inspect** = identity + provenance logging, **Redirect** = node relabeling, **Fabricate** = synthetic node/edge injection.

Consistency requirement: $D_{\mathrm{FS}}(\mathcal{TG})$ must satisfy all invariants (I1–I11, G1–G7). Since D_{FS} preserves the tree-like structure, $D_{\mathrm{FS}}(\mathcal{TG})$ also has bounded treewidth, and consistency constraints are checkable as MSO queries in $O(n)$.

10 Verification Plan

Table 2: Four-layer verification plan with effort estimates.

Layer	Tool	Properties	Effort	Confidence
1	TLA+/TLC	Graph invariants, DPO rules, GTXN protocol	2–3 wk	Model-checked
2	SMT (Z3)	Crash refinement, WAL correctness	3–4 wk	Push-button
3	Rust types	State machine transitions, lifetime safety	4–6 wk	Compile-time
4	Coq/Perennial	Atomicity, unforgeability, treewidth	6–12 mo	Mechanized proof

Layer 1 is complete: four TLA+ specifications in `formal/` model the core graph (`sotfs_graph.tla`), GTXN protocol (`sotfs_transactions.tla`), capability delegation (`sotfs_capabilities.tla`), and crash recovery (`sotfs_crash.tla`).

11 Conclusion and Future Work

SOTFS demonstrates that treating filesystem metadata as an explicit typed graph, with operations formalized as algebraic DPO rewriting rules, unlocks structural properties—bounded treewidth, discrete curvature monitoring, and capability-gated projections—that are impossible in conventional implicit-graph designs.

The immediate next steps are: (Phase 2) FUSE prototype on Linux implementing the type graph and DPO engine as Rust crates; (Phase 3) integration with SOTOS as a native filesystem server; (Phase 4) incremental formal verification; (Phase 5) structural differentiators (treewidth enforcement, curvature monitoring, deceptive projections).

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